5.13.16

AI25BTECH11030 -Sarvesh Tamgade

Question:Let k be an integer such that the triangle with vertices

$$\mathbf{A} = \begin{pmatrix} k \\ -3k \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ k \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -k \\ 2 \end{pmatrix}$$

has area 28 sq. units. Find the orthocenter of the triangle.

Solution:

$$\text{Area} = \frac{1}{2} |\mathbf{M}| = 28 \tag{0.1}$$

$$\text{where} \quad \mathbf{M} = \begin{pmatrix} \mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} \end{pmatrix} = \begin{pmatrix} 5 - k & -2k \\ 4k & 2 + 3k \end{pmatrix} \tag{0.2}$$

$$\det(\mathbf{M}) = (5 - k)(2 + 3k) - (-2k)(4k) = 10 + 13k + 5k^2 \tag{0.3}$$

$$10 + 13k + 5k^2 = \pm 56 \tag{0.4}$$

After solving,

$$k = 2 \tag{0.5}$$

$$\text{For } k = 2, \quad \mathbf{A} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \tag{0.6}$$

The equation of the altitude through $\mathbf{A}$ is:

$$\mathbf{x}^T \mathbf{n} = 1 \tag{0.7}$$

This altitude passes through $\mathbf{A}$, so

$$\mathbf{A}^T \mathbf{n} = 1 \tag{0.8}$$

$$\begin{pmatrix} 2 & -6 \end{pmatrix} \mathbf{n} = 1 \tag{0.9}$$

and the direction vector $\mathbf{n}$ of the altitude is perpendicular to $\mathbf{B} - \mathbf{C}$, so

$$(\mathbf{B} - \mathbf{C})^T \mathbf{n} = 0 \tag{0.10}$$

$$\begin{pmatrix} 7 & 0 \end{pmatrix} \mathbf{n} = 0 \tag{0.11}$$

The augmented matrix is:

$$\left(\begin{array}{cc|c} 7 & 0 & 0 \\ 2 & -6 & 1 \end{array}\right) \tag{0.12}$$

Row reduction steps:

First, $R_2 \to R_2 - \frac{2}{7}R_1$:

$$\left(\begin{array}{cc|c} 7 & 0 & 0 \\ 0 & -6 & 1 \end{array}\right) \tag{0.13}$$

$$\mathbf{n} = \begin{pmatrix} 0 \\ -\frac{1}{6} \end{pmatrix} \tag{0.14}$$

The equation of the altitude through $\mathbf{B}$ is:

$$\mathbf{x}^T \mathbf{n} = 1 \tag{0.15}$$

This altitude passes through $\mathbf{B}$, so

$$\mathbf{B}^T \mathbf{n} = 1 \tag{0.16}$$

$$\begin{pmatrix} 5 & 2 \end{pmatrix} \mathbf{n} = 1 \tag{0.17}$$

and the direction vector $\mathbf{n}$ of the altitude is perpendicular to $\mathbf{A} - \mathbf{C}$, so

$$(\mathbf{A} - \mathbf{C})^T \mathbf{n} = 0 \tag{0.18}$$

$$\begin{pmatrix} 4 & -8 \end{pmatrix} \mathbf{n} = 0 \tag{0.19}$$

The augmented matrix is:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 4 & -8 & 0 \end{array}\right) \tag{0.20}$$

Row reduction steps: First, $R_2 \to R_2 - \frac{4}{5}R_1$:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 0 & -\frac{48}{5} & -\frac{4}{5} \end{array}\right) \tag{0.21}$$

Next, $R_2 \to -\frac{5}{48}R_2$:

$$\left(\begin{array}{cc|c} 5 & 2 & 1 \\ 0 & 1 & \frac{1}{12} \end{array}\right) \tag{0.22}$$

Back substitute: $R_1 \to R_1 - 2R_2$

$$\left(\begin{array}{cc|c} 5 & 0 & \frac{5}{6} \\ 0 & 1 & \frac{1}{12} \end{array}\right) \tag{0.23}$$

$$\mathbf{n} = \begin{pmatrix} \frac{1}{6} \\ \frac{1}{12} \end{pmatrix} \tag{0.24}$$

First altitude:

$$\begin{pmatrix} 0 & -\frac{1}{6} \end{pmatrix} \mathbf{x} = 1 \tag{0.25}$$

Second altitude:

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{12} \end{pmatrix} \mathbf{x} = 1 \tag{0.26}$$

Combining both equations:

$$\begin{pmatrix} 0 & -\frac{1}{6} \\ \frac{1}{6} & \frac{1}{12} \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{0.27}$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} \frac{1}{6} & \frac{1}{12} & 1 \\ 0 & -\frac{1}{6} & 1 \end{array}\right) \tag{0.28}$$

Perform the row operation $R_1 \to R_1 + \frac{1}{2}R_2$:

$$\left(\begin{array}{cc|c} \frac{1}{6} & 0 & \frac{3}{2} \\ 0 & -\frac{1}{6} & 1 \end{array}\right) \tag{0.29}$$

Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} 9 \\ -6 \end{pmatrix} \tag{0.30}$$

Answer: The orthocentre of the triangle is

$$\boxed{\mathbf{x} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}}$$

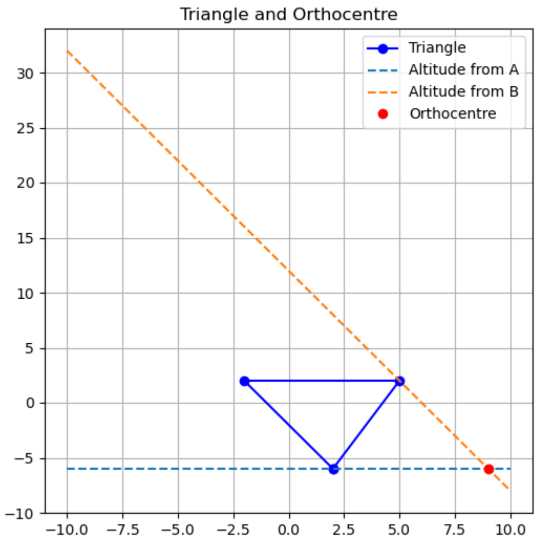

Fig. 0.1: Vector Representation